

BanglaBlend Based Evaluation of Bengali Formal and Colloquial Text Classification

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Abstract—Bangla, one of the most widely spoken languages globally, exhibits a significant diglossic nature characterized by two primary stylistic forms: Sadhu Bhasha (formal/classical) and Cholit Bhasha (colloquial). With the proliferation of digital communication, the blending of these styles often leads to grammatical inconsistency and challenges in computational processing. This study presents a comprehensive comparative analysis for the stylistic classification of Bengali text using the BanglaBlend dataset, comprising 7,350 balanced instances. A diverse range of models is evaluated across three categories: traditional Machine Learning (ML), Deep Learning (DL), and Transformer-based Natural Language Processing (NLP) architectures. Our experimental results indicate a clear performance hierarchy, where Transformer models significantly outperform traditional approaches. Specifically, XLM-RoBERTa achieved the highest performance with an accuracy of 94% and an F1-score of 0.94, closely followed by Bangla-BERT at 93.4%. Deep Learning models like CNN and BiLSTM also showed robust results, reaching 92% accuracy, whereas traditional ML models plateaued at 81%. These findings underscore the necessity of contextualized embeddings and self-attention mechanisms in capturing the nuanced morphological and syntactic patterns of the Bangla language, providing a robust foundation for downstream tasks such as stylistic normalization, content moderation, and sentiment analysis.

Index Terms—Bangla NLP, Style Classification, Sadhu Bhasha, Cholit Bhasha, Transformer Models, XLM-RoBERTa, Deep Learning.

I. INTRODUCTION

Bengali, or Bangla, is one of the richest languages in the world, with over 230 million speakers, primarily in Bangladesh and India, in terms of its literary, cultural, and historical heritage [1], [2]. This language is characterized linguistically by two most significant stylistic forms: Cholit Bhasha, the colloquial form used as a rule in everyday spoken and written communication, and Sadhu Bhasha, which is of a more elevated and highly formal variety, often applied in literary and official contexts [3]. This dichotomy has long been a serious consideration for the linguistics of the Bengali language, given its importance in maintaining stylistic coherence and grammatical structure. [4]. With modern times, especially in digital and online communications, these two stylistic forms are often mixed together within one sentence, resulting in the loss of stylistic coherence, grammatical irregularity, and semantic ambiguity [5]. As Bengali text very often relies

on subtle semantic clues, negation patterns, and contextual dependencies difficult to model automatically, such stylistic merging makes the task of computational processing even more challenging [6]. Its status as a low-resource language, where there are still few large-scale, annotated, and style-aware datasets, makes these difficulties even more acute [7]. In fact, classical machine learning techniques like Support Vector Machines, Decision Trees, Random Forests were primarily adopted for initial studies on Bangla text classification, leveraging manually designed features including TF-IDF, n-grams [8]. Although these performed relatively well, those found it difficult to identify the subtle stylistic characteristics and long-range dependencies in Bengali texts [9]. Recent advances in deep learning techniques make it possible for models such as Long Short-Term Memory (LSTM), BiLSTM, Convolutional Neural Networks (CNN), etc., to effectively identify sequential & contextual features in Bangla text classification tasks [10]. Furthermore, leveraging contextual representations obtained through comprehensive multilingual datasets, transformer models such as multilingual BERT, ELECTRA, etc., achieved impressive feats in classifying Bangla sentences & documents [8]. Compared to classical and deep learning solutions, these approaches prove particularly advantageous in low-resource environments, where fully optimized multilingual transformers can potentially attain better-performing outputs [11].

This paper aims to employ the Bangla Blend corpus for developing a robust learning platform for effectively classifying Bengali texts in terms of Sadhu Bhasha (Sanskrit) & Cholit Bhasha (colloquial). This paper aims to identify how well transformer models in NLP, Deep learning, classical machine learning perform in effectively deciphering these subtle stylistic & contextual aspects in Bangali linguistics. This paper aims to assess multilingual & language-specific transformer models for precise feasibility in computing Bengali linguistics better. In reality, supporting real-world applications in content filtering, opinion mining, etc., this paper aims to showcase a comprehensive comparative analysis in highlighting optimal solutions for precise & contextual classifying in Bangla texts.

II. RELATED WORK

For low-resource languages like Bangla, recent developments in transformer-based Natural Language Processing (NLP) have greatly enhanced text classification performance. Despite this, in real-world scenarios, model performance is still constrained due to issues with informal language, semantic ambiguity, and stylistic variation. Fatema et al. studied transformer-based sentiment analysis for Bangla social media content with models like BanglaBERT and multilingual BERT [12]. Their tests show the competitiveness of BanglaBERT but its moderate performance while processing informal and context-rich Bangla text, with a micro-F1 score of around 0.71. This study illustrates the fact that transformer-based NLP models, with their strengths, do not always achieve high accuracy in noisy social media environments. The transformer models for Bangla emergency post classification on social media platforms were explored by Nabil et al. [13]. Even with heavy mixing of languages, casual wording, and uneven data, their top model reached about 85% accuracy, showing how well transformer designs could do when facing challenges arising from messy, real-world inputs. Bhowmick et al. leveraged transformer models-like multilingual BERT and XLM-RoBERTa-to investigate the estimation of sentiment in Bangla [14]. Their work relied on these deep structures to elicit the emotional sentiment expressed in the language. According to their research, transformer-based NLP models outperform traditional machine learning-based methods. However, across various datasets, the accuracy remained less than 90% for many real-world scenarios. This variability can be attributed to the impact of different characteristics of datasets on transformer performance. Sourav et al. assessed transformer-based models on a unified Bangla emotion corpus in a multi-class classification context [15]. Their findings indicate that while contextual embeddings help improve the categorization of emotional content, the average accuracy remained below 90%, identifying the intrinsic challenge of multi-class text classification in Bangla based on intricate linguistic and emotional expressions. Furthermore, transformer-based techniques for identifying hate speech and cyberbullying in Bangla social media text were studied by Emon et al. [16]. With an accuracy of roughly 85%, their refined transformer models showed successful semantic modeling and highlighted the degradation of performance brought on by informal language and stylistic inconsistency. Transformer-based NLP techniques for Bangla text classification generally achieve accuracy in the range of roughly 71% to 88%, depending on task complexity and data quality, according to the reviewed studies. Despite these developments, not much research has been done expressly on stylistic categorization, specifically differentiating Sadhu and Cholit Bangla. The current study's evaluation of transformer-based and neural approaches for Bangla stylistic differentiation is motivated by this gap.

III. PROPOSED METHODOLOGY

The BanglaBlend dataset is used to classify Bengali Sadhu and Cholit linguistic forms in a methodical manner. The

entire workflow of our work is depicted in the figure.

A. Dataset Description

The BanglaBlend dataset, a comprehensive corpus of 7,350 annotated sentences carefully created to support research in the stylistic variations of the Bengali language, is used to assess the performance of our suggested model. Sadhu Bhasha (Saint/Formal) and Cholit Bhasha (Common/Colloquial) are the two main linguistic forms that are the subject of this dataset. With 3,675 instances for Class 0 (Sadhu) and 3,675 instances for Class 1 (Cholito), the dataset maintains a perfectly balanced distribution to guarantee a strong training environment and reduce class bias. The BanglaBlendCleanedData.xlsx file, which offers the data in a preprocessed format after thorough normalization and noise removal covering a wide range of contexts including literature, news, and social dialogues, was used for the study's experimental phase. As a result, in the field of Bengali computational linguistics, this dataset functions as a specialized resource for a variety of natural language processing (NLP) tasks, including style classification and automatic language translation. Some samples from the dataset is given in Figure 1

Sentence	Labels(English)	Labels(Bangla)	Class
শুনিয়া অতান্ত পুলকিত হইলাম	Saint	Sadhu(সাধু)	0
বিদেশ আমার সখল ফুরাইয়া গিয়াছে	Saint	Sadhu(সাধু)	0
সুমন আমাকে খাবার দিয়েছিল	Common	Cholito(চলিত)	1
ফুটবলে এটিকে বলা হয় কিলার বল বা ফাইনাল বল	Common	Cholito(চলিত)	1
কুমা আমার মাকে সব বলেছিল	Common	Cholito(চলিত)	1
সাহসে কুলাচ্ছে না	Common	Cholito(চলিত)	1
শিলা রবিনকে জ্বলে নিয়া গিয়াছিল	Saint	Sadhu(সাধু)	0
তাহার গায়ের রঙ যেন রঙ নেহে তাহা আভা	Saint	Sadhu(সাধু)	0
সকলেই বাইরে হইয়া রাস্তায় একধারে আসিয়া দাড়াইল	Saint	Sadhu(সাধু)	0

Fig. 1. Data Sample

B. Data Preprocessing

The preprocessing pipeline differs across the three methodological tracks (ML, DL, and NLP) to ensure optimal model performance.

1) *Machine Learning Preprocessing*: Cleaning and vectorization were carried out on the text data for traditional ML models. This involved removing duplicates, special symbols, emojis, URLs, punctuation, and changing all text to lowercase. Further, the cleaned text was tokenized, and stop words were removed. Finally, TF-IDF vectorization was applied to the preprocessed text corpus to transform text features into numerical feature representations suitable for ML classifiers.

2) *For Deep Learning Preprocessing*: Data preprocessing for DL-based models involved token-level representations. Text cleaning was done, as in the ML stage, followed by tokenization and sequence padding to arrive at the same input length. Embedding matrices were created using pre-trained Word2Vec or FastText embeddings for Bengali and allowed the deep learning architectures to understand and make use of the semantic relationships existing among words.

3) *For Natural Language Processing Preprocessing*: The transformer-based language models had minimal preprocessing to maintain contextual integrity. Text normalization consisted of replacing unusual characters, normalizing whitespace, and employing a model-specific tokenizer such as Bangla BERT, IndicBERT, or XLM-RoBERTa on a subword level.

C. Data Splitting

70% of the dataset used for training, 15% for testing, and 15% for validation subsets were generated using a stratified split in order to preserve the class distribution.

IV. MODEL DESCRIPTION

To comprehensively evaluate performance, experiments have been conducted using Machine Learning, Deep Learning, and NLP-based Transformer models.

A. NLP Transformer Models

1) *Bangla-BERT*: Bangla-BERT [17] is a transformer-based model pre trained on large Bengali text corpora, using self-attention mechanisms to learn contextual relations between sentences. It was fine-tuned on the ALERT dataset for multi-class classification.

2) *IndicBERT*: IndicBERT is a multilingual model based on the ALBERT framework and supports several Indian languages, including Bengali, with the intention of applying parameter sharing to reduce computational cost without much performance compromise in low-resource settings.

3) *XLM-RoBERTa*: XLM-RoBERTa, trained on 100 languages that include Bengali, is fine-tuned with adding a classification layer to the encoder output.

4) *mBERT*: Multilingual BERT (mBERT) is Google's original multilingual BERT, pre-trained on Wikipedia texts across 104 languages, including Bengali, using the standard BERT architecture and the Masked Language Modeling (MLM) objective. It uses a single WordPiece vocabulary across all languages, without language-specific tokenization.

B. Machine Learning Models

1) *XGBoost*: XGBoost represents an ensemble learning algorithm based on gradient-boosted decision trees [18], which was chosen because of scalability and efficiency with sparse TF-IDF features.

2) *LightGBM*: Another gradient-boosting framework is LightGBM, which uses histogram algorithms to accelerate the training and reduce memory consumption, and thus it efficiently deals with a large text feature space while achieving accuracy comparable to XGBoost.

3) *Random Forest*: The Random Forest classifier constructs multiple decision trees and combines their results through majority voting, which makes it robust against noise and overfitting.

4) *CatBoost*: CatBoost is a gradient boosting algorithm that builds decision trees sequentially, where each new tree corrects the residual errors of the previous ensemble while efficiently handling categorical features and reducing prediction bias.

C. Deep Learning Models

1) *Convolutional Neural Network (CNN)*: As part of Deep Learning, CNN is used to learn local and spatial patterns in text data using convolutional filters by identifying the most crucial n-gram features for classification.

2) *Long Short-Term Memory (LSTM)*: Then, LSTM networks overcome the limitation of RNN, due to the use of the gating mechanism, allowing the modeling of long-range dependencies to contextualize better religiously aggressive content.

3) *Bidirectional LSTM (BiLSTM)*: BiLSTM extends the architecture of LSTM by processing the text sequence in both forward and backward directions. The bidirectional mechanism helps capture contextual information from previous and following words, enhancing semantic understanding.

4) *Artificial Neural Network (ANN)*: ANN is a computational model inspired by the human brain that learns complex patterns by adjusting weighted connections between interconnected layers of neurons through iterative error minimization.

V. RESULT AND DISCUSSION

This section deals with the results procured from traditional machine learning approaches, deep learning approaches, and transformer models for NLP for the binary classification problem conducted on the chosen Bengali text dataset. Accuracy, precision, recall, and F1 score are some of the standard parameters used to evaluate the performance of all models described in this chapter. Results demonstrate the crucial role of learning context and semantics for understanding and processing the language of Bengali by demonstrating increasing performance from traditional machine learning approaches to deep learning models and finally to transformer models for NLP. Table I summarizes the overall classification performance for all evaluated models.

A. Machine Learning (ML) Results

The baseline models used for comparison are the conventional machine learning models LightGBM, CatBoost, Random Forest, and XGBoost. The models are performing comparatively well, though the accuracy lies in the range 0.79 to 0.81. Among the models having the highest accuracy of 0.81 and macro-F1 score of 0.81, CatBoost is performing better than all the models closely followed by the model XGBoost and then the model Random Forest.

Each type of machine learning model follows almost the same pattern in its class-wise evaluation, in which class 0 follows greater precision, whereas class 1 follows relatively greater recall values. This pattern indicates that machine learning models work better in detecting obvious lexical patterns but lack capabilities to generalize and understand more subtle linguistic expressions. The poor performance can be attributed to its tendency to follow shallow hand-crafted or statistical features, which do not generalize well to represent longer-range dependencies in the semantics and syntax of the Bengali language. Even if machine learning models are more efficient in computations, those still lack in dealing with more complex

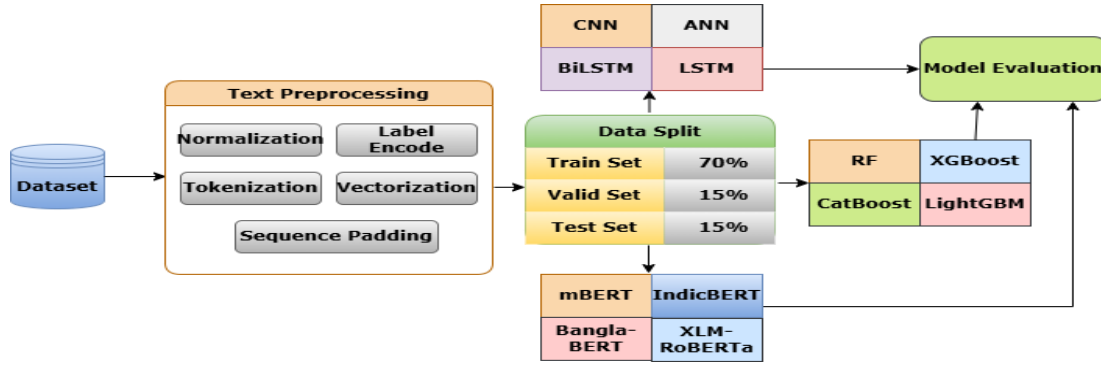


Fig. 2. Methodology Workflow

linguistic processing like understanding heavy semantics of language, as presented in Bengali.

TABLE I
OVERALL CLASSIFICATION PERFORMANCE ACROSS ML, DL, AND NLP MODELS

Model	Category	Precision	Recall	F1-Score	Accuracy
Bangla-BERT	NLP	0.94	0.93	0.93	0.93
XLM-RoBERTa		0.95	0.94	0.94	0.94
Indic-BERT		0.87	0.86	0.86	0.86
m-BERT		0.89	0.88	0.88	0.88
LightGBM	ML	0.79	0.79	0.79	0.79
CatBoost		0.82	0.81	0.81	0.81
Random Forest		0.80	0.80	0.80	0.80
XGBoost		0.81	0.80	0.80	0.80
ANN	DL	0.91	0.90	0.90	0.90
CNN		0.92	0.92	0.92	0.92
BiLSTM		0.92	0.92	0.92	0.92
LSTM		0.89	0.89	0.89	0.89

B. Deep Learning (DL) Results

To overcome the limitations of traditional ML approaches, ANN, CNN, and BiLSTM were some of the deep learning models that were evaluated. The models were used to show the effectiveness of learning representations using neural networks for text-classification tasks because all of them performed better than all other baselines.

The macro F1-score of 0.90 and accuracy of 0.90 were an outstanding improvement over ML. In addition, there was a misalignment in terms of class recall, where class 1 had relatively higher recall compared to class 0. This suggests that although models based on ANN are able to handle non-linear interactions of features, those do not incorporate or handle sequential patterns.

With an accuracy of 0.92 and a macro F1-score of 0.92, it was observed that the CNN and BiLSTM models exhibited the best results among all deep learning models. The capability of CNN to explore discriminative local n-gram patterns was established by observing well-balanced precision and recall

values for both classes. BiLSTM's capability to explore bidirectional dependencies among text was utilized to improve the understanding of context, which led to well-balanced class performance. Although the results were impressive, it was observed that there was a plateau effect at 92% accuracy.

C. Transformer-Based NLP Results

In this case, transformer models like Bangla BERT, XLM-RoBERTa, Indic BERT, and mBERT were found to be always better than the ML model and DL model. This can be attributed to the capabilities of these models in handling deep semantic aspects using a self-attention mechanism through pre-training.

Bangla BERT was very effective in learning representations specific to the language with a good balance of performance on both classes, obtaining an accuracy of 0.934 and a macro F1-score of 0.934. XLM-RoBERTa was slightly better than Bangla BERT, obtaining an accuracy of 0.94 and a macro F1-score of 0.94. Its precision on class 0 and high value of recall on class 1 resulted in its outstanding ability for cross-lingual and contextual feature acquisition.

Although Indic-BERT performed well, with an accuracy of 0.8605 and macro-F1 of 0.8599, it was less accurate than that of Bangla-BERT and XLM-RoBERTa. This difference in accuracy shows that although being pre-trained for multiple languages is advantageous for Indic-BERT, some linguistic nuances of Bengali may not necessarily be captured. mBERT performed better than Indic-BERT with an accuracy of 0.8812 and macro-F1 of 0.8809, but still less accurately than the other models.

Overall, the transformer models had better generalization capabilities and were more robust, especially when dealing with ANVs, compared to the conventional ML and DL models.

D. Comparative Analysis

Among the assessed model families, a distinct performance hierarchy became apparent:

ML Models(0.80) < DL Models(0.92) < NLP Models(0.94)

This sequence illustrates how feature-engineered methods give way to representation learning and, ultimately, context-aware language modeling. DL models use equential semantics, transformer models use self-attention mechanisms to incorporate bidirectional contextual information, and machine learning models rely on surface-level textual features. Bangla-BERT and XLM-RoBERTa’s superior performance highlights the value of pretrained transformer models for Bengali NLP tasks and shows that those are appropriate for real-world applications that demand high accuracy and robustness.

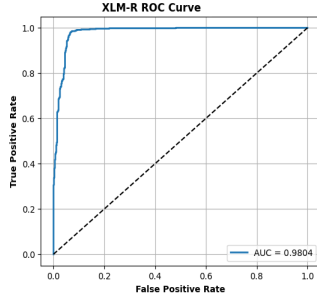


Fig. 3. ROC curves (XLM-RoBERTa)

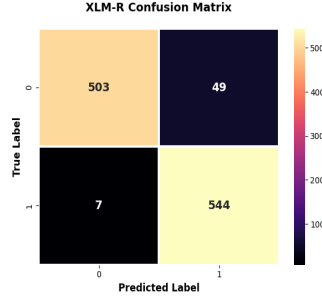


Fig. 4. Confusion matrix (XLM-RoBERTa).

E. Discussion

The results of the experiment unequivocally show that deep contextualised language understanding, as opposed to relying solely on surface-level lexical features, is necessary for effective Bengali text classification. Transformer-based NLP models demonstrated superiority in capturing semantic subtleties, long-range dependencies, and implicit linguistic cues found in Bengali social media text, outperforming both traditional machine learning and deep learning approaches in every experiment.

With an accuracy and macro F1-score of 0.94, XLM-RoBERTa outperformed the other transformer models evaluated, closely followed by Bangla-BERT with an accuracy of 0.934. Large-scale pretraining and selfattention mechanisms, which allow the learning of both global and local contextual representations, are responsible for these models’ excellent performance. While XLM-RoBERTa uses strong cross-lingual representations to improve generalisation across a variety of linguistic patterns, Bangla-BERT gains from language-specific pretraining, which makes it extremely effective at capturing Bengali morphology and syntax. Conversely, despite being multilingual, Indic-BERT and mBERT showed relatively poorer performance, indicating that more extensive multilingual coverage might weaken Bengali’s language-specific representational ability.

With accuracies of about 0.80, conventional machine learning models like LightGBM, CatBoost, Random Forest, and XGBoost performed moderately. Semantic ambiguity and contextual dependencies are difficult for these models to represent because those mainly rely on handcrafted or superficial textual features. Although those are interpretable and computationally

efficient, their efficacy for complex language understanding tasks is limited by their incapacity to capture minute contextual variations.

With accuracies as high as 0.92, deep learning models such as ANN, CNN, and BiLSTM significantly outperformed ML baselines. By identifying local patterns and sequential dependencies, CNN and BiLSTM models showed robust and balanced class-wise performance. Nevertheless, despite these advancements, transformers’ explicit global context modelling is still absent from DL models, creating a performance ceiling that transformer-based methods are able to surmount. The confusion matrix and ROC curve displayed in Figures 3 and 2, respectively, provide additional evidence of XLMRoBERTa’s efficacy. Strong diagonal dominance in the confusion matrix indicates balanced, accurate classification across both classes with little misclassification. Bengali social media text is inherently complex, as evidenced by the majority of errors occurring in samples with implicit or context-dependent linguistic patterns. Strong separability between classes is also shown by the ROC curve, which stays near the top-left corner and indicates a high true positive rate and low false positive rate. When taken as a whole, these visual analyses verify that XLM-RoBERTa demonstrates stable and dependable decision-making behaviour in addition to achieving superior quantitative metrics.

Transformer models achieve a better balance between precision and recall for both classes, indicating robust generalisation and less bias towards any one class, according to analysis of class-wise metrics. Implicit and context-dependent expressions, where subtle semantics, pragmatic intent, or ambiguous phrasing make discrimination difficult even for sophisticated models, are mostly responsible for the remaining misclassifications.

Overall, the findings demonstrate that transformer-based architectures are especially well-suited for Bengali text classification tasks, particularly in informal and low-resource social media contexts. Those are very useful for practical applications like content moderation, opinion mining, and harmful content detection because of their capacity to represent complex semantics and contextual relationships. This gives future socially conscious NLP systems a solid foundation.

VI. CONCLUSION

In order to distinguish between the formal Sadhu Bhasha and the informal Cholito Bhasha, this study tackled the problem of stylistic classification in the Bengali language. By utilizing sequential representation learning, deep learning architectures in particular, CNN and BiLSTM offered significant improvements. Transformer-based models produced the biggest performance gains, and the most successful architectures were XLM-RoBERTa and Bangla-BERT. XLM-RoBERTa’s superior accuracy of 94% demonstrates the effectiveness of self-attention mechanisms and extensive pretraining in simulating low-resource languages with high stylistic variability. Our results indicate that language-specific or robust cross-lingual transformers are the best option for tasks

TABLE II
CLASS-WISE PRECISION, RECALL, F1-SCORE, AND OVERALL
ACCURACY OF ML, DL, AND NLP MODELS

Model	Class	Precision	Recall	F1-Score	Accuracy
Bangla-BERT	Shadhu	0.96	0.91	0.93	0.93
	Cholito	0.91	0.96	0.94	
XLM-RoBERTa	Shadhu	0.99	0.90	0.94	0.94
	Cholito	0.91	0.99	0.95	
Indic-BERT	Shadhu	0.92	0.79	0.85	0.86
	Cholito	0.82	0.93	0.87	
m-BERT	Shadhu	0.93	0.82	0.87	0.88
	Cholito	0.84	0.94	0.89	
LightGBM	Shadhu	0.80	0.78	0.79	0.79
	Cholito	0.78	0.81	0.80	
CatBoost	Shadhu	0.85	0.76	0.80	0.81
	Cholito	0.78	0.86	0.82	
Random Forest	Shadhu	0.82	0.76	0.79	0.80
	Cholito	0.78	0.84	0.81	
XGBoost	Shadhu	0.84	0.75	0.79	0.80
	Cholito	0.77	0.86	0.81	
ANN	Shadhu	0.96	0.83	0.89	0.90
	Cholito	0.85	0.97	0.91	
CNN	Shadhu	0.92	0.92	0.92	0.92
	Cholito	0.92	0.92	0.92	
BiLSTM	Shadhu	0.93	0.91	0.92	0.92
	Cholito	0.91	0.93	0.92	
LSTM	Shadhu	0.90	0.87	0.88	0.89
	Cholito	0.88	0.91	0.89	

requiring a deep comprehension of Bengali linguistic nuances. Expanding this framework to handle "style-mixed" sentences sentences that contain both Sadhu and Cholito forms will be the main goal of future work. Furthermore, the incorporation of Explainable AI (XAI) methods is intended to be investigated in order to gain a better understanding of the particular linguistic markers, such as pronoun variations and verb inflections, that influence the decision-making of these transformer-based models..

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